

PIATRA INSTITUTE

Hadamard Matrix at Order 668

Verified defect-1920 cyclic Goethals-Seidel candidate and exact radius-4 local certificate

Resolution status	UNRESOLVED
Best exact GS defect score	1920 (target: 0)
Public checkpoint verified	2496
Research date	21 July 2026

This document does not claim a Hadamard matrix of order 668.

It preserves the strongest exact computational result obtained in the available execution environment, together with reproducible verification code and the precise remaining gap.

Prepared from the supplied Piatra Institute research prompt, status snapshot 20 July 2026.

1. Executive verdict

No complete construction or nonexistence proof was obtained.

The order-668 problem remains open in this work. The supplied acceptance criterion requires either a 668×668 sign matrix H with $H H^T = 668 I$, or a complete proof that none exists. The best candidate found here fails that criterion: its maximum off-diagonal Gram entry is 8, not 0.

The strongest verified advance is a cyclic Goethals-Seidel (GS) candidate represented by four $+1/-1$ sequences of length 167. Under the exact periodic-autocorrelation defect metric used by the public GS/SDS search lane, its score is 1920, improving the latest public checkpoint I could verify, 2496, by 576 points or 23.08 percent. This is a computational improvement, not a solution and not a proof of a global record over unpublished work.

Quantity	Verified value
Order implied by the GS array	668
Sequence dimensions	4×167
Row sums	(17, 17, 9, 3)
SDS parameters	(167; 75, 75, 79, 82; 144)
Exact defect score E	1920
Required score for a solution	0
Maximum combined autocorrelation	8
Nonzero independent shifts	39 of 83
Matrix-level status	Not Hadamard

2. Exact constructive reduction

Let a_1, a_2, a_3, a_4 be sign sequences indexed by $\mathbb{Z}_{167}$. For each nonzero shift t , define the combined periodic autocorrelation

$$C(t) = \sum_{r=1}^4 \sum_{i=0}^{166} a_r(i) a_r(i+t \bmod 167).$$

Let A, B, C and D be the 167×167 circulant matrices whose first rows are the four sequences, and let R be the back-diagonal reversal matrix. The standard GS array is

$$H = \begin{bmatrix} A & B R & C R & D R \\ -B R A & D^T R & -C^T R & \\ -C R -D^T R & A & B^T R & \\ -D R & C^T R & -B^T R & A \end{bmatrix}.$$

Circulant commutativity cancels the off-diagonal block products. The diagonal block is $A A^T + B B^T + C C^T + D D^T$, whose first row is $(668, C(1), \dots, C(166))$. Therefore this expanded 668×668 matrix is Hadamard if and only if

$$C(t) = 0 \text{ for every } t = 1, \dots, 166.$$

This is a sufficient construction family, not a characterization of all Hadamard matrices of order 668. Failure inside cyclic GS cannot prove global nonexistence.

2.1 Row-sum constraint and the ten signatures

At Fourier frequency zero, the row sums $s_r = \sum_i a_r(i)$ must satisfy

$$s_1^2 + s_2^2 + s_3^2 + s_4^2 = 668.$$

Each s_r is odd because the sequence length is 167. Up to signs and permutation, the complete set of nonnegative sorted solutions is:

ID	Sorted absolute row-sum signature
1	(1, 1, 15, 21)
2	(1, 9, 15, 19)
3	(3, 3, 5, 25)
4	(3, 3, 11, 23)
5	(3, 3, 17, 19)
6	(3, 7, 9, 23)
7	(3, 7, 13, 21)
8	(3, 9, 17, 17)
9	(5, 9, 11, 21)
10	(7, 13, 15, 15)

The strongest candidate uses the equivalent ordered signature (17, 17, 9, 3). Its negative-set sizes are $k_r = (167 - s_r)/2 = (75, 75, 79, 82)$.

2.2 Supplementary-difference-set form

Let X_r be the positions of -1 in sequence r , and let $N_r(t)$ count ordered differences $x-y=t$ inside X_r . For the active signature, $C(t)=0$ is equivalent to

$$N_1(t) + N_2(t) + N_3(t) + N_4(t) = 144 \text{ for every nonzero } t \text{ in } \mathbb{Z}_{167}.$$

Thus an exact solution in this lane is an SDS with parameters (167; 75,75,79,82;144).

3. Exact objective and candidate defect

The search and all acceptance checks use integer arithmetic. Define

$$E = \sum_{t=1}^{166} C(t)^2.$$

Because $C(t)=C(167-t)$, only 83 shifts are independent. In this fixed-sum family every $C(t)$ is divisible by 4. Writing $d_t=C(t)/4$ gives

$$E = 32 \sum_{t=1}^{83} d_t^2.$$

For the archived candidate, $\sum d_t^2=60$, hence $E=1920$. The 83 independent shifts have the following exact distribution:

Combined $C(t)$	Number of independent shifts	Contribution to $\sum d_t^2$
0	44	0
+4	19	19
-4	13	13
+8	2	8
-8	5	20
Total nonzero	39	60

The complete shift-by-shift vector is in candidate/combined_periodic_autocorrelation.csv. The exact remaining gap is therefore 39 nonzero equations: 32 have magnitude 4 and 7 have magnitude 8.

4. Search portfolio executed

The computation did not rely on one local heuristic alone. The following materially different lanes were implemented or audited. Scores are directly comparable only within the exact GS periodic-defect metric shown above.

Lane	Best exact score	Interpretation
Public cyclic GS/SDS checkpoint	2496	Verified and used as the starting certificate
Coupled fixed-weight tabu and multi-swap search	1920	Best result; active candidate
Second signature (3,3,11,23)	1920	Same score, but 45 nonzero shifts
Reflected projection, fixed row sums	2688	Fourier/time-domain projection
Reflected projection, free row sums	2464	Different basin; no exact candidate
Twin/repeated-sequence GS family	3072	Lower-dimensional special family
Paley-deletion family	6848	Delete 8,8,4,1 points from Paley blocks
Random-phase Fourier target moves	1920	10,000 proposals; did not beat seed
Bounded exact MILP repair	No score <1920	One smaller neighborhood proved infeasible; wider run timed out

The search code for the principal lanes is included under search/. The public 2496 sequence is included and reverified under the same verifier, avoiding a metric mismatch.

5. Independent verification and exact local certificate

5.1 Two independent sequence verifiers

A pure-Python verifier and an independently written C++17 verifier both parse exactly four rows of 167 entries, reject non-sign tokens, compute all 166 periodic shifts, and agree on:

Check	Python	C++
Shape	4 x 167	4 x 167
Row sums	17,17,9,3	17,17,9,3
Full score E	1920	1920
Maximum C(t)	8	8
Nonzero independent shifts	39	39
Exact complementary quad	false	false

5.2 Exhaustive fixed-weight Hamming-radius-4 audit

The candidate is not merely a one-swap local minimum. Every sign state that preserves all four row sums and lies at Hamming distance at most 4 was enumerated exactly. These are precisely: one balanced swap in one row; two balanced swaps in distinct rows; or a balanced four-bit change in one row.

Neighborhood class	States checked	Best neighbor score	Improving state?
Hamming distance 2: one swap	27,722	2368	No
Hamming distance 4: one swap in each of two rows	288,189,040	2688	No
Hamming distance 4: two swaps in one row	46,882,338	2048	No
Total	335,099,100	2048	No

Conclusion: under the fixed row-sum signature, the score-1920 candidate is a strict local minimum throughout Hamming radius 4. The nearest better target must cross a score barrier or require at least 6 bit changes.

This statement is exact for the enumerated fixed-sum neighborhood. It does not imply global optimality within the signature and does not address other construction families.

5.3 Matrix-level expansion

The four sequences were expanded into the full 668 x 668 GS sign matrix and $H H^T$ was computed in exact 64-bit integer arithmetic. This direct audit produced:

Matrix-level quantity	Value
Dimensions	668 x 668
All entries +1/-1	Yes
Every diagonal Gram entry	668
Maximum absolute off-diagonal Gram entry	8
Nonzero ordered off-diagonal Gram entries	52,104
Squared Frobenius norm of $H H^T - 668 I$	1,282,560
Hadamard identity satisfied	No

The direct matrix calculation agrees with the sequence reduction and prevents a false positive caused by a block-construction convention or reversal error.

6. Bounded exact repair

A mixed-integer linear model was constructed around the score-1920 state. Flip variables enforce exact row-sum balance. Pair variables linearize the quadratic effect of jointly flipped endpoints. One-hot variables encode every post-move defect d_t in an exact bounded integer range, and the model asks whether $\sum d_t^2$ can fall below 60.

Model	Selected endpoints	Hamming cap	Target	Result
Certified bounded neighborhood	48 total (12 per row)	8	$\sum d_t^2 \leq 59$	Infeasible, HiGHS status 8
Wider exploratory neighborhood	96 total (24 per row)	10	$\sum d_t^2 \leq 59$	No feasible solution before 25 s limit; not a proof

The first result excludes improvement only within the encoded endpoint subset and Hamming cap. It is not a certificate of optimality for the full cyclic GS search space.

7. What survives scrutiny

- The four-sequence reduction is mathematically sufficient: a zero-defect quad would expand to an exact order-668 Hadamard matrix.
- The row-sum signature and SDS parameters are internally consistent and independently verified.
- The score-1920 artifact is parseable, exact, and reproducible; Python, C++, and full-matrix checks agree.
- The improvement from the public score-2496 checkpoint to 1920 is exact under the same metric.
- The fixed-sum radius-4 local-minimum certificate is exhaustive over 335,099,100 states.

8. What remains unresolved

- The candidate still has 39 nonzero independent autocorrelation equations and therefore does not construct a Hadamard matrix.
- No argument shows that an order-668 Hadamard matrix must be cyclic GS, GS of any kind, cocyclic, group-developed, or contained in any tested ansatz.
- No exhaustive search over all fixed-sum sequences, all ten signatures, or all Hadamard equivalence classes was completed.
- The local and bounded exact certificates do not prove that 1920 is the minimum cyclic GS defect.
- No nonexistence proof was found.

9. Rational continuation

The exact radius-4 certificate changes the correct next step. Continuing ordinary one- or two-swap descent from this state is mathematically pointless. The next computational campaign should use one of the following barrier-crossing routes:

1. Run a full population-based PatternBoost campaign on GPU, seeded by the entire symmetry orbit of the 1920 candidate but with free row sums during improvement. The 2026 transformer work reports that large GS instances benefit from nonlocal phase moves, parallel tempering, and learned proposal distributions.
2. Build exact Hamming shells 6, 8, 10, ... with column generation: add endpoints based on dual violations, retain all 83 defect equations, and record solver certificates. The small fixed endpoint model is already implemented in the archive.
3. Move beyond cyclic GS: search noncyclic abelian groups, cocyclic matrices, orbit matrices, and switching trades. A cyclic failure cannot settle order 668.
4. Exploit the two score-1920 signatures jointly. Canonicalize under the full residual GS symmetry group, compare invariants, and use the distinct basins as crossovers rather than repeatedly restarting one basin.

10. Reproducibility package

The accompanying ZIP archive is designed so that the numerical claims can be checked without trusting this report. From the extracted root, run:

```
./reproduce.sh
```

The script runs the Python sequence verifier, the one-swap audit, the full 668 x 668 matrix audit, compiles the independent C++ verifier, and compares the reported metrics. The complete radius-4 exhaustive results and source code are also preserved under verification/.

Path	Purpose
<code>candidate/best_GS_sequences_score1920.csv</code>	Central four-sequence artifact
<code>candidate/combined_periodic_autocorrelation.csv</code>	All 166 exact shift defects
<code>verification/verify_sequences.py</code>	Strict pure-Python verifier
<code>verification/verify_sequences.cpp</code>	Independent C++ verifier
<code>verification/fixed_weight_hamming_radius4_audit.json</code>	Consolidated exact local certificate
<code>verification/matrix_level_audit.py</code>	Full GS expansion and exact Gram audit
<code>search/experiment_manifest.json</code>	Machine-readable research summary
<code>MANIFEST.sha256</code>	SHA-256 for every archived file

11. Sources and status checks

- [1] Piatra Institute, Prompt for Resolving the Hadamard Matrix Problem at Order 668, status snapshot 20 July 2026. User-supplied PDF.
- [2] Epoch AI, Hadamard Matrices, FrontierMath open problem page, accessed 21 July 2026.
- [3] OEIS A019442, Numbers m such that a Hadamard matrix of order m exists, update dated 30 May 2026.
- [4] G. Williamson, O. Yacobi, P. Zinn-Justin, Generating Hadamard matrices with transformers, arXiv:2604.11101, April 2026.
- [5] renaissancefieldlite/Hadamard_Proof public repository, order-668 GS/SDS checkpoint 2496, accessed 21 July 2026.
- [6] D. Z. Djokovic and I. S. Kotsireas, Goethals-Seidel difference families with symmetric or skew base blocks, 2018.

Status statement: public sources checked on 21 July 2026 still describe 668 as the smallest order for which no real Hadamard matrix is known. This report makes no claim about unpublished results.

Appendix A. Nonzero independent defect shifts

t	C(t)	d_t	t	C(t)	d_t	t	C(t)	d_t
2	-4	-1	3	-4	-1	5	4	1
7	-4	-1	8	-4	-1	9	-4	-1
10	4	1	11	4	1	15	4	1
16	8	2	21	-4	-1	22	-8	-2
23	4	1	24	4	1	30	4	1
32	-4	-1	36	4	1	37	4	1
39	-8	-2	40	8	2	41	-8	-2
42	-4	-1	45	-4	-1	47	4	1
52	4	1	55	4	1	57	4	1
58	4	1	59	4	1	60	4	1
61	-4	-1	62	4	1	63	4	1
64	-8	-2	70	-4	-1	73	4	1
75	-4	-1	79	-4	-1	83	-8	-2

There are 39 nonzero shifts among $t=1,\dots,83$. The remaining shifts follow $C(167-t)=C(t)$.

Appendix B. Acceptance condition

The archived candidate becomes a genuine solution only if a future search produces four sequences for which every entry in candidate/combined_periodic_autocorrelation.csv is zero. At that point, the GS expansion script must generate a 668 x 668 sign matrix and two independent exact verifiers must confirm $H H^T = 668 I$ entry by entry. The current artifact deliberately fails that gate and is labeled accordingly.